

The efficacy of multiple versus single hyaluronic acid injections: a systematic review and meta-analysis.

PMID: 29268731 | Indexed in PubMed

Intra-articular hyaluronic acid (IA-HA) is a common therapy used to treat knee pain and suppress knee inflammation in knee osteoarthritis (OA), typically prescribed in regimens ranging from a single injection to 5 weekly injections given once weekly. We conducted a systematic review to determine the efficacy of IA-HA, with subgroup analyses to explore the differences in knee pain and adverse events (AEs) across different dosing regimens. We conducted a systematic search of the literature to identify studies evaluating IA-HA for the management of knee OA compared to IA-saline. Primary outcome measure was the mean knee pain score at 13 Weeks (3 months) or 26 weeks (6 months). Secondary outcome was the number of treatment-related AEs and treatment-related serious adverse events (SAEs). We evaluated differences in levels of pain and AEs/SAEs between dosing regimens compared to IA-Saline. Thirty articles were included. Overall, IA-HA injections were associated with less knee pain compared to IA-Saline injections for all dosing regimens. 2-4 injections of IA-HA vs. IA-Saline produced the largest effect size at both 3-months and 6-months (Standard mean difference [SMD] = -0.76; -0.98 to -0.53, 95% CI, $P < 0.00001$, and SMD = -0.36; -0.63 to -0.09 95% CI, $P = 0.008$, respectively). Additionally, single injection studies yielded a non-significant treatment effect at 3 and 6 months, while ≥ 5 injections demonstrated a significant improvement in pain only at 6 months. Five or more injections of IA-HA were associated with a higher risk of treatment-related AEs compared to IA-Saline (Risk ratio [RR] = 1.67; 1.09 to 2.56 95% CI, $p = 0.02$), which was a result not seen within the 1 and 2-4 injection subgroups. Overall, 2-4 and ≥ 5 injection regimens provided pain relief over IA-Saline, while single injection did not. Intra-articular injections of HA used in a 2-4 injection treatment regimen provided the greatest benefit when compared to IA-Saline with respect to pain improvement in patients with knee OA, and was generally deemed safe with few to no treatment-related AEs reported across studies. Future research is needed to directly compare these treatment regimens.

Is hyaluronate sodium effective in the management of knee osteoarthritis? A placebo-controlled double-blind study.

PMID: 20467406 | Indexed in PubMed

The aim of this study was to investigate the effect of intra-articular hyaluronic acid (HA) injection on pain and function in knee osteoarthritis (OA). Forty-eight patients with knee OA were included in this study. The patients were randomized into two groups: one group received HA injections (average molecular weight [MW] 1.5 million Da), and the other group received placebo containing 0.9% saline. Three injections of HA or placebo were given at weeks 1, 2 and 3. The evaluation instruments were: Visual Analog Scale (VAS); Likert Scale; Lequesne Index; the Western Ontario and McMaster Universities (WOMAC) Index for Osteoarthritis pain, stiffness, and function, and WOMAC pain subgroups (pain on walking, climbing stairs, at night, on sitting and lying down, on standing); the number of analgesics taken; changes in knee flexion angle; and patient satisfaction. Assessment was performed at weeks 1, 3, 5, and 14 after the first injection. Significant improvement for almost all parameters was noted in both groups ($P < 0.05$), except for WOMAC pain subscore on walking at final assessment (week 14) which showed greater improvement in the HA-treated group (35.2% versus 9.1%; $P = 0.01$). HA treatment was effective in the management of knee OA and improved knee pain and functional outcome, but there was no statistically significant difference in functional and symptom improvement with respect to saline

(placebo) injection.

Efficacy of intra-articular sodium hyaluronate in the treatment of knee osteoarthritis.

PMID: 12740675 | Indexed in PubMed

To assess the efficacy of intra-articular hyaluronic acid in patients with knee osteoarthritis, sixty female patients with knee osteoarthritis were randomised to three weekly intra-articular injections of 30 mg sodium hyaluronate (Na HA) with a high molecular weight (1.0 to 2.9 million Da) or 40 mg 6-methylprednisolone acetate (6-MPA). The clinical assessments included pain at rest, at weight-bearing and on walking, Lequesne Index and active range of knee flexion. Assessments were done at baseline, at week 4, and at months 3 and 6. A significant decrease in VAS scores for pain at rest, at weight-bearing and pain on walking, and in Lequesne index was found in both groups at week 4 when compared to baseline and there was no significant differences between the two groups. However, at 3(rd) month improvement in all pain scores and Lequesne index was found in favour of hyaluronic acid. At 6(th), no significant difference was found between the treatment groups. Improvement in pain was accompanied by an increase in joint flexion at week 4 and at month 3 in both groups. Both treatments were well-tolerated. The results showed that both intra-articular hyaluronic acid and 6-MPA treatments provide clinically significant improvement and demonstrated that Na HA has a long-term beneficial effect in patients with knee osteoarthritis.

Functional and perceived response to intra-articular hyaluronan injection in patients with knee osteoarthritis: persistence of treatment effects over 5 months.

PMID: 19296086 | Indexed in PubMed

Intra-articular injections of hyaluronic acid are currently indicated as a symptom-modifying treatment for knee osteoarthritis to palliate symptoms and improve function. A series of three to five weekly injections are typically delivered once per week and a period of at least 6 months must pass between consecutive injection series. Little is known about why some patients do not respond to this treatment or about its effects on outcome measures recorded during functional activities. In addition to investigating short-term self-report and measured functional changes, the purpose of this study was to evaluate the persistence of treatment effects over the typical time interval between consecutive treatment series. According to the a priori definition for response to treatment, over two-thirds of patients experienced improved knee function during the typical period between series of intra-articular injection of hyaluronic acid while a one-third did not demonstrate a change in self-report scores. Differences in response were similarly seen between groups in functional tests. Many had immediate benefits and although some did not retain their gains, others experienced a delayed response in the following months. Knee flexion range of motion was related to self-report and tested function, and improvements in range of motion were associated with concurrent improvements in other functional measures across the testing period. A short functional test and knee-specific questionnaire may provide an index that provides an indication as to whether or not a particular patient is likely to respond to hyaluronan injections. Such information may provide valuable input when decisions are made regarding which treatment option is recommended to individuals with knee osteoarthritis.

Physical therapy, intra-articular dextrose prolotherapy, botulinum neurotoxin, and hyaluronic acid for knee osteoarthritis: randomized clinical trial.

PMID: 32776763 | Indexed in PubMed

This study aimed to compare the efficacy of four treatments in the management of knee osteoarthritis. We carried out a randomized clinical trial with four study arms in an outpatient Department of Physical Medicine and Rehabilitation at a University Hospital. In total, 120 patients with knee osteoarthritis ≥ 50 years of age were randomly allocated to four groups. The primary outcome was knee pain in visual analog scale and the secondary outcome was the Knee Injury and Osteoarthritis Outcome Score. The exercise was prescribed daily for all participants throughout the study. For physical therapy (group 1), participants received superficial heat, transcutaneous electrical nerve stimulation and pulsed ultrasound. We administered a single intra-articular injection of botulinum neurotoxin type A (group 2) and three injections of hyaluronic acid (group 3) or 20% dextrose (group 4) to patients in the corresponding groups. Mixed analysis of variance showed that there was statistically significant difference between the groups in pain ($P < 0.001$), and Knee Injury and Osteoarthritis Outcome Score ($P < 0.001$). Pairwise between- and within-group comparisons showed that botulinum neurotoxin and dextrose prolotherapy were the most, and hyaluronic acid was the least efficient treatments for controlling pain and recovering function in patients. An intra-articular injection of botulinum toxin type A or dextrose prolotherapy is effective first-line treatments. In the next place stands physical therapy particularly if the patient is not willing to continue regular exercise programs. Our study was not very supportive of intra-articular injection of hyaluronic acid as an effective treatment of knee osteoarthritis.

Editorial Commentary: Knee Hyaluronic Acid Viscosupplementation Reduces Osteoarthritis Pain.

PMID: 26433240 | Indexed in PubMed

In contrast to the AAOS knee osteoarthritis guidelines, systematic review of overlapping meta-analyses shows that viscosupplementation with intra-articular hyaluronic acid injection reduces knee osteoarthritis pain and improves function according to the highest level of evidence.

Therapeutic effects of hyaluronic acid on osteoarthritis of the knee. A meta-analysis of randomized controlled trials.

PMID: 14996880 | Indexed in PubMed

The magnitude of the therapeutic effects of intra-articular injection of hyaluronic acid on osteoarthritis of the knee is still in question. The aim of this meta-analysis was to elucidate the therapeutic efficacy and safety of intra-articular injection of hyaluronic acid for osteoarthritis of the knee. We conducted a meta-analysis of twenty blinded randomized controlled trials that compared the therapeutic effect of intra-articular injection of hyaluronic acid with that of intra-articular injection of a placebo to treat osteoarthritis of the knee. The outcome end points were classified into three categories: pain with activities, pain without activities, and function. The outcome measures of the efficacy of hyaluronic acid were the mean differences in the efficacy scores between the hyaluronic acid and placebo groups. The outcome measure of the safety of hyaluronic acid was the relative risk of adverse events. Intra-articular

injection of hyaluronic acid can decrease symptoms of osteoarthritis of the knee. We found significant improvements in pain and functional outcomes with few adverse events. However, there was significant between-study heterogeneity in the estimates of the efficacy of hyaluronic acid. Subgroup analysis and meta-regression analysis showed that lower methodological quality such as a single-blind or single-center design resulted in higher estimates of hyaluronic acid efficacy, that introduction of acetaminophen as an escape analgesic in the trial resulted in lower estimates of hyaluronic acid efficacy, and that patients older than sixty-five years of age and those with the most advanced radiographic stage of osteoarthritis (complete loss of the joint space) were less likely to benefit from intra-articular injection of hyaluronic acid. This meta-analysis confirmed the therapeutic efficacy and safety of intra-articular injection of hyaluronic acid for the treatment of osteoarthritis of the knee. Additional well-designed randomized controlled trials with high methodological quality are needed to resolve the continued uncertainty about the therapeutic effects of different types of hyaluronic acid products on osteoarthritis of the knee in various clinical situations and patient populations. Therapeutic study, Level II-3b (systematic review; nonhomogeneous Level-I studies). See Instructions to Authors for a complete description of levels of evidence.

Intrarticular treatment of osteoarthropathy knee with polynucleotides: a pilot study with medium-term follow-up.

PMID: 23830403 | Indexed in PubMed

Knee osteoarthritis is a major cause of disability in the elderly. Many therapies are nowadays available, ranging from non-pharmacologic to pharmacological approaches like visco-supplementation, oral supplements or topical treatments, but a flawless treatment is still to be found. Visco-supplementation represents a valid treatment option for reducing pain associated with knee osteoarthritis and improving function in the affected joint. Many literature data report on the efficacy and safety profiles of hyaluronic acid in knee osteoarthritis, however the efficacy of intra-articular hyaluronic acid remains controversial, in fact while several clinical trials claimed a disease-modifying effect for hyaluronic acid, subsequent meta-analyses have cast doubts on this fact. The ideal intra-articular treatment for osteoarthritis should not only provide a mechanical protection of the cartilage surface, but also restore chondrocytes homeostasis by restoring the physiological articular micro-environment and supplying nutrients. In this perspective an innovative medical product made up of polynucleotides (Condrotide) has been developed. The aim of this study is to test the 2-months efficacy in pain relief and improving function of intra-articular injections of Condrotide in patients with knee osteoarthritis or with grade III or IV chondropathy. Ninety-five subjects (33 men, 62 women), aged between 53 and 80, were included between May 2011 to July 2012. All subjects received intra-articular injections of Condrotide and were evaluated with the Knee injury and Osteoarthritis Outcome Score (KOOS), the NRS scale for pain assessment, the measurement of the range of motion (R.O.M.). In all subjects a significant improvement was found in KOOS score after 60 days. The mean global NRS pain decreased in both groups and there was also a R.O.M. improvement. These results show that the intra-articular administration of nucleotides in subjects with both severe knee arthritis and chondropathy can be recommended since is able to reverse in the short and medium term symptoms and function with a significant improvement in quality of life.

Clinical therapy of hyaluronic acid combined with platelet-rich plasma for the treatment

of knee osteoarthritis.

PMID: 30186448 | Indexed in PubMed

Knee osteoarthritis is the most common degenerative disease of the joints caused by articular cartilage injury, degradation of the joint edge and subchondral bone hyperplasia. Various treatments are used to alleviate the symptoms of patients with knee osteoarthritis, including analgesics and intra-articular injections. Platelet-rich plasma (PRP) is an autologous and multifunctional platelet concentrate of the blood, which stimulates the cartilage healing process and improves the damage caused by articular disease. Hyaluronic acid (HA) is an effective treatment for patients with knee osteoarthritis. In the current study, the effectiveness of PRP and HA combination therapy administered via intra-articular injections for patients with knee osteoarthritis was analyzed. A total of 360 patients with knee osteoarthritis were randomized into four different treatment groups as follows: Double-blind treatment with PRP (2-14 ml); double-blind treatment with HA (0.1-0.3 mg); combination therapy of PRP and HA; and placebo groups. Following treatment, all patients were evaluated using the Western Ontario and McMaster Universities Arthritis Index (WOMAC) and Common Toxicity Criteria. The most common treatment-emergent adverse events were hypertension and proteinuria. The current study demonstrated that PRP and HA treatment significantly improved arthralgia, and PRP treatment was determined to be significantly more effective than HA treatment using the WOMAC pain score ($P < 0.05$). PRP and HA combination treatment significantly improved arthralgia, reduced humoral and cellular immune responses and promoted angiogenesis, which improved the patients' histological parameters compared with PRP or HA treatment alone. These results suggested that PRP and HA combination treatment may be a potential treatment option for patients with knee osteoarthritis in the future.

Joint and Soft Tissue Injections.

PMID: 37590853 | Indexed in PubMed

Joint and soft tissue injections can be divided into two primary categories: diagnostic and therapeutic. Diagnostic injections facilitate a diagnosis by using a local anesthetic to identify the site of pain or through fluid aspiration for analysis. Therapeutic injections are categorized by the type of injectate and include corticosteroids, hyaluronic acid, dextrose prolotherapy, and platelet-rich plasma. Diagnostic and therapeutic injections are most accurate under direct visualization. Corticosteroid injections help treat adhesive capsulitis and tenosynovitis but are not recommended for intratendinous injections. Hyaluronic acid has limited benefits for knee osteoarthritis. Dextrose prolotherapy injections treat tendinopathy and degenerative joint pain. Platelet-rich plasma injections effectively treat common extensor tendinopathy and knee arthritis; however, the evidence does not support its use for other soft tissue injuries. Preparation for injections includes patient education, consent, proper patient positioning, and obtaining the necessary supplies. Local infection, fractures, and allergy to injection substrates are contraindications to joint and soft tissue injections. Potential complications include pain, swelling, and redness. Corticosteroid injections into soft tissue may cause atrophy and depigmentation, and repeated injections can cause cartilage and tendon degeneration. Optimizing conservative, noninjection treatments, such as oral and topical analgesics, activity modification, or rehabilitation, is also important.